Nivi Jha

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Education

Jaypee University of Information Technology
Bachelor of Technology – Computer Science & Engineering (CGPA: 8.5/10) 2023 – 2027

Experience

Summer Intern | IIT Delhi, India June 2025 – July 2025

- Automated Linux system monitoring with cron-scheduled Bash scripts using AWK and grep to collect CPU, memory, disk, and network metrics every 60 seconds, trigger threshold-based alerts, and compress logs automatically.
- Built a Python security-monitoring agent that forwarded structured JSON audit logs for 100 daily SSH login events with under 2 seconds latency.

Projects

MedTracker | Intelligent Healthcare Record Management System GitHub | Live

Tech: Next.js, React.js, Node.js/Express, MongoDB, JWT + OAuth 2.0, Cloudinary, Multer, Redis

- Engineered a multi-tier healthcare records platform (Next.js client, Express REST API with 36 endpoints, MongoDB with 7 models) with JWT and Google OAuth 2.0 authentication.
- Designed a cache-aside layer with Redis (30-day TTL) in front of MongoDB as the source of truth, so AI report analysis runs only once per record. Implemented automatic failover from LLaMA to Gemini.
- Wrote 21 automated tests (Vitest, Supertest, in-memory MongoDB) and integrated them into a GitHub Actions CI/CD pipeline that gates releases on passing tests.

CardioVision | Coronary Artery Stenosis Detection using Deep Learning GitHub | Live

Tech: React.js, Python, FastAPI, NumPy, Gemini AI

- Built and deployed a full-stack medical-imaging application (React client, FastAPI backend) that runs YOLOv8 inference to detect and segment coronary stenosis in angiograms.
- Benchmarked 3 YOLOv8 model variants and served them through a 5-endpoint API for detection, segmentation, and heatmap overlays.
- Implemented and tested a geometry-based classifier that calculates percentage stenosis from reference and minimum lumen diameters and maps results to five severity levels.

Chat Constellation | Graph-Based Conversational Interaction Visualization GitHub | Live

Tech: React.js, Node.js/Express, MongoDB, NVIDIA LLM API, React Flow, JWT + Google OAuth 2.0, Docker

- Designed a full-stack conversation-visualization app: React client with 3 views (chat, constellation graph, galaxy), Express REST API (10 endpoints), MongoDB data model.
- Implemented authentication with JWT (7-day expiry), bcrypt password hashing and Google OAuth 2.0.
- Containerized 3 services (MongoDB, Express, Nginx) with Docker Compose; integrated the NVIDIA LLM API to generate bilingual English/Hindi responses.

Technical Skills

Languages: Java, Python, JavaScript, SQL
Core CS: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks
Frontend & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs
Databases: MySQL, MongoDB, Redis
Systems: Docker, Linux, Git, GitHub
Cloud: AWS Lambda, AWS Comprehend, Vercel, Render

Achievements

- **Coding Profiles** — Solved 300+ data structures and algorithms problems on LeetCode (Java). (**LeetCode: vinj21**)
- **AWS Academy Graduate – Cloud Foundations (2025)** — cloud computing and AWS fundamentals.
- **Hacktoberfest 2025 contributor** — open source collaboration via Git and GitHub pull requests.
- **Web Dev Team Lead, JYC** — Led a team of 5 developers and mentored 3 juniors on React.js and Git version-control workflows.